UGED1111 Logic

Symbolic Language
and
Truth Table II

CHAN, Hei Man

Logical Relations and Properties of Propositions

Logical Properties of propositions

1. Tautologous
2. Self-contradictory
3. Contingent

Logical relations of propositions:

1. Logically Equivalent
2. Contradictory
3. Consistent
4. Inconsistent

Properties of Propositions

<u>**Tautologous**</u>

- **Necessarily true**
- **it is true regardless of the truth values of its components**
- **True in every possible interpretation , not just in actual world**
- **E.g.** $P \vee \sim P$

Properties of Propositions

<u>Self-contradictory</u>

- Necessarily false
- it is false regardless of the truth values of its components
- false in every logical possibility , not just in actual world
- E.g. $P \cdot \sim P$

Properties of Propositions

Contingent

- Its truth value varies depending on the truth values of its components.
- True in some logical possibilities; false in some logical possibilities
- i.e. at least one line true and at least one line false
- E.g. $\mathbf{P \cdot Q}$

Discussion 3

What is the logical property of the following propositions?

1. $(P \vee Q) \equiv \sim Q$
2. $(P \supset Q) \cdot (\sim Q \cdot P)$

Answer 3.1

P	Q	(P	∨	Q)	≡	~	Q
T	T	T	T	T	F	F	T
T	F	T	T	F	T	T	F
F	T	F	T	T	F	F	T
F	F	F	F	F	F	T	F

Contingent

Answer 3.2

P	Q	(P	⊃	Q)	·	(~	Q	·	P)
T	T	T	T	T	F	F	T	F	T
T	F	T	F	F	F	T	F	T	T
F	T	F	T	T	F	F	T	F	F
F	F	F	T	F	F	T	F	F	F

Self-contradictory

Relations of Propositions

Logically Equivalent

For any wffs α and β, α and β are logically equivalent just in case they have the same truth value in every logical possibility.

P	Q	~	P	∨	Q	/	P	⊃	Q
T	T	F	T	T	T		T	T	T
T	F	F	T	F	F		T	F	F
F	T	T	F	T	T		F	T	T
F	F	T	F	T	F		F	T	F

Relations of Propositions

Contradictory

For any wffs α and β, α and β are contradictory just in case α and β have different truth values in every logical possibility.

P	Q	~	Q	·	P	/	P	⊃	Q
T	T	F	T	F	T		T	T	T
T	F	T	F	T	T		T	F	F
F	T	F	T	F	F		F	T	T
F	F	T	F	F	F		F	T	F

Relations of Propositions

Consistent

For any wffs α and β, α and β are consistent just in case there is at least one logical possibility that α and β are both true.

P	Q	~	Q	v	P	/	P	⊃	Q
T	T	F	T	T	T		T	T	T
T	F	T	F	T	T		T	F	F
F	T	F	T	F	F		F	T	T
F	F	T	F	T	F		F	T	F

Relations of Propositions

Inconsistent

For any wffs α and β, α and β are inconsistent just in case there is no logical possibility that α and β are both true.

P	Q	~	Q	≡	P	/	P	·	Q
T	T	F	T	F	T		T	T	T
T	F	T	F	T	T		T	F	F
F	T	F	T	T	F		F	F	T
F	F	T	F	F	F		F	F	F

Discussion 4

What is/are the logical relations of the following propositions?

1. $P \supset Q$ / $\sim Q \cdot P$
2. $P \vee (Q \supset S)$ / $S \vee (Q \supset P)$

Answer

P	Q	(P	⊃	Q)	/	(~	Q	·	P)
T	T	T	T	T		F	T	F	T
T	F	T	F	F		T	F	T	T
F	T	F	T	T		F	T	F	F
F	F	F	T	F		T	F	F	F

Contradictory and inconsistent

****If two propositions are Contradictory, they must be inconsistent.**

Answer

P	Q	S	P ∨ (Q ⊃ S) / S ∨ (Q ⊃ P)
T	T	T	
T	T	F	
T	F	T	
T	F	F	
F	T	T	
F	T	F	
F	F	T	
F	F	F	

Answer

P	Q	S	P	∨	(Q	⊃	S)	/	S	∨	(Q	⊃	P)
T	T	T	T	T	T	T	T		T	T	T	T	T
T	T	F	T	T	T	F	F		F	T	T	T	T
T	F	T	T	T	F	T	T		T	T	F	T	T
T	F	F	T	T	F	T	F		F	T	F	T	T
F	T	T	F	T	T	T	T		T	T	T	F	F
F	T	F	F	F	T	F	F		F	F	T	F	F
F	F	T	F	T	F	T	T		T	T	F	T	F
F	F	F	F	T	F	T	F		F	T	F	T	F

Logically Equivalent and consistent

Validity

Checking the validity of an argument with truth table:

1. In a truth table for argument, we need to **list out all the possible truth values of premises and conclusion in the truth table**.
2. To clearly indicate premises and conclusion, we use **" / "** to separate the premises and **"// "** to separate the last premise and the conclusion.
3. If you can find **any rows showing all true premises and false conclusion** , then the argument is invalid. (Indicate that row when you find it.)

Discussion 4

Use a complete truth table to check whether the argument is valid.

1. $P \supset Q$ / $\sim P$ // Q

Answer

P	Q	P	⊃	Q	/	~	P	//	Q
T	T	T	T	T		F	T		T
T	F	T	F	F		F	T		F
F	T	F	T	T		T	F		T
F	F	F	T	F		T	F		F

Invalid

Discussion 5

Use a complete truth table to check whether the argument is valid.

1. $\sim T \supset S \ / \ \sim S \vee (\sim T \supset R) \ / \ \sim T \ // \ R$

Answer

R	S	T	~	T	⊃	S	/	~	S	∨	(~	T	⊃	R)	/	~	T	//	R
T	T	T	F	T	T	T		F	T	T	F	T	T	T		F	T		T
T	T	F	T	F	T	T		F	T	T	T	F	T	T		T	F		T
T	F	T	F	T	T	F		T	F	T	F	T	T	T		F	T		T
T	F	F	T	F	F	F		T	F	T	T	F	T	T		T	F		T
F	T	T	F	T	T	T		F	T	T	F	T	T	F		F	T		F
F	T	F	T	F	T	T		F	T	F	T	F	F	F		T	F		F
F	F	T	F	T	T	F		T	F	T	F	T	T	F		F	T		F
F	F	F	T	F	F	F		T	F	T	T	F	F	F		T	F		F

Valid

Case

$$(N \cdot P \equiv (Q \vee R)) \supset (\sim S \cdot M) \;/\; (N \cdot P) \;/\; Q \;//\; (\sim S \cdot M)$$

Simple proposition as component: 6

We have to list 2^6 (i.e. 64) rows!

Indirect truth table saves our time!

Indirect Truth Table

Indirect truth tables provide a faster way for checking validity of arguments.

◆The idea is to check whether the argument is possible to have all true premises and a false conclusion by making assumptions.

◆If it is possible for the argument to have all true premises and a false conclusion , then the argument is invalid. Otherwise, valid.

Indirect Truth Table

We need to first **assume the conclusion is false and the premises are true**(which means it is invalid).
Then we have to see if there is any **contradiction** in the computation.

- If there is a contradiction, then the assumption is false , i.e. the argument is valid.

- If there is **no** contradiction, then there is a case that the conclusion is false and the premises are true, i.e. the argument is invalid.

Example

Whether the following argument is valid?

~(P ∨ R)/Q ≡ ~P//Q ·~ R

~	(P	∨	R)	/	Q	≡	~	P	//	Q	·	~	R

Case 1 : $\sim(P \vee R)/Q \equiv \sim P//Q \cdot \sim R$

- If there is a contradiction, then the assumption is false , i.e. the argument is valid.
- If there is **no** contradiction, then there is a case that the conclusion is false and the premises are true, i.e. the argument is invalid.

- In the indirect truth table of this argument ,there is a contradiction.
- Indicate the contradiction by circling it.
- It is valid!!!!

~	(P	v	R)	/	Q	≡	~	P	//	Q	·	~	R
T	F	F	F		F	T	F	T		F	F	T	F
1	3	2	3		7	1	8	9		6	1	5	4

Indirect Truth Table

(P	≡	Q)	⊃	S	/	~	S	·	P	//	P	v	Q
F	T	F	T	F		T	F	T	T		F	F	F

Similar techniques you may have used before

SUDOKU!!!!!

4	8			6	1			2
		2	3					1
1	9				7	6	4	5
	1	6		2		5		9
3		7	1		8	4		6
9		4		7		3	1	
2	3	1	8				5	7
7					2	1		
5			7	1			6	3

Indirect Truth Table

- Sometimes we need to make further assumption if the truth value of certain letters are not fixed by previous steps.

P	⊃	Q	/	Q	⊃	P	//	P	≡	Q
T	T	F		F	T	T		T	F	F
3	1	4		4	1	3		2	1	2
F	T	T		T	T	F		F	F	T
3	1	4		4	1	3		2	1	2

- If no contradiction is found, then the argument is invalid, you can stop further computation.
- If a contradiction has been found, you CANNOT stop the computation. **You need to exhaust every logical possibility**.

Reference

- Copi, I., Cohen, C., & McMahon, K. (2011). *Introduction to Logic (14th ed. / Irving M. Copi, Carl Cohen, Kenneth McMahon ed.). Upper Saddle River, NJ: Pearson Education.*
- Hurley, P. (2015). *A Concise Introduction to Logic (12th ed.). Australia ; Stamford, Ct.: Cengage Learning.*